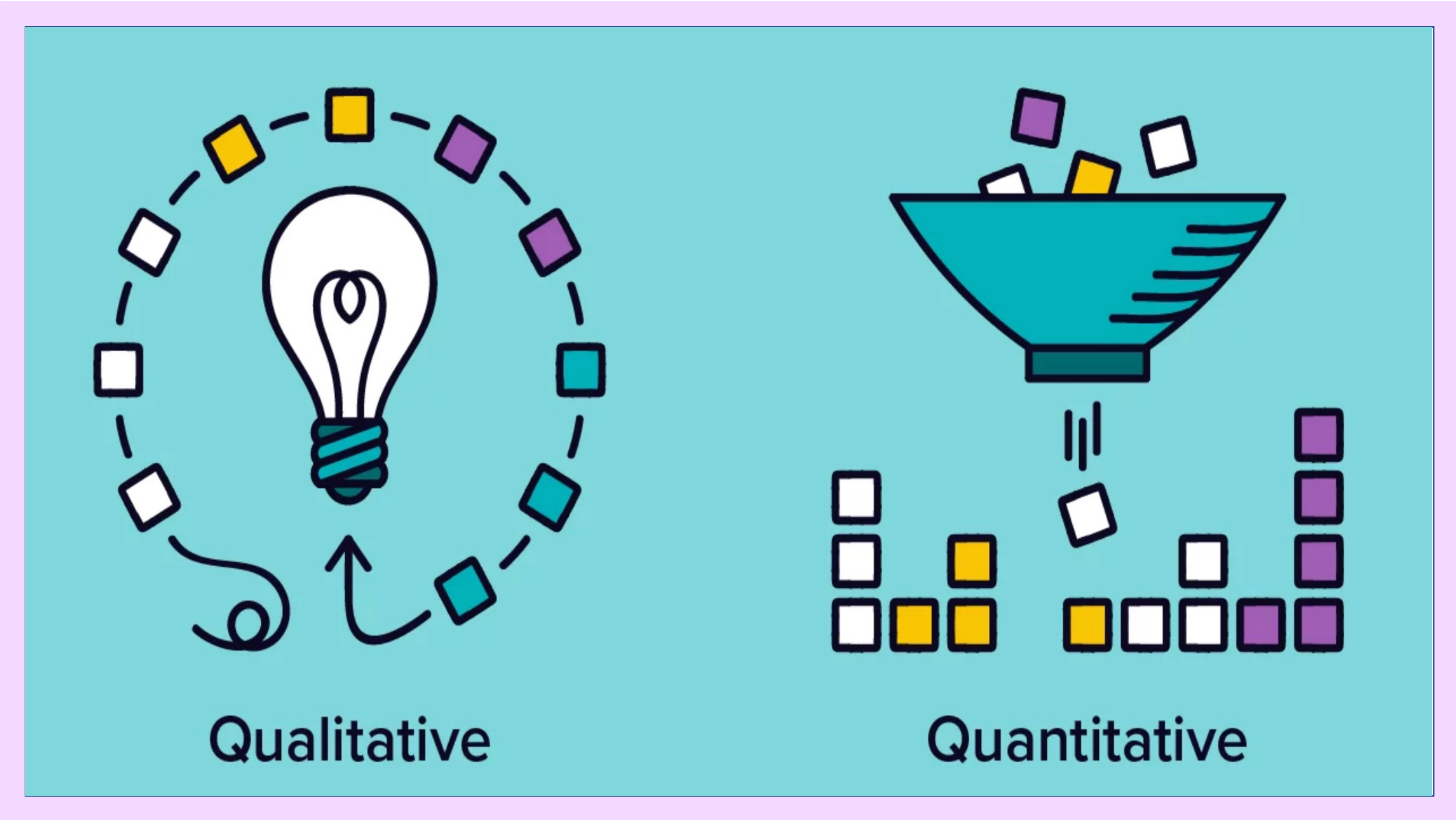
Qualitative
Quantitative

I can...

- ✓ identify quantitative observations, explain how they differ from qualitative observations, and use both appropriately to support scientific inquiry. [1.1.4]
- ✓ provide examples of observations that are relevant to a scientific investigation and explain how a specific observation might address a testable question. [1.1.5]

Vocabulary

Think & Talk

1. What kind of data do you use in daily life?
2. At what steps in the Scientific Method is data useful?
3. How is data useful to us as scientists?

Qualitative vs Quantitative Data

Qualitative Data	Quantitative Data
Descriptions	Numbers
Data can be observed but not directly measured	Data which can be measured
Colors, textures, smells, tastes, appearance, beauty, value judgements, perceptions, etc.	Length, height, area, volume, mass, speed, time, temperature, humidity, cost, percentages, ages, etc.
Qualitative → Quality	Quantitative → Quantity

Practice: Macchiato

Qualitative Observations	Quantitative Observations
frothy	61℃
dark	355 mL
glass cup	36 元
bitter	477g
good smell	8% milk by volume
delicious	

In your science journal:

Imagine we are experimenting to answer the question "How does soil acidity affect the growth of Ficus plants?"

1. What qualitative observations may be useful and why?

2. What quantitative observations may be useful and why?

Measure what is measurable, and make measurable what is not so.

~ Galileo Galilei

- What did Galileo mean by this?
- Why would this advice help scientists?
- How can scientists today follow his advice?